

# Capstone Project I: Mutual Fund Analytics

## Day 1 Report — Project Setup + Data Ingestion (ETL)

**Report Date:** June 23, 2026

**Git Repository:** [github.com/anveshtammineni/mutual-fund-analytics](https://github.com/anveshtammineni/mutual-fund-analytics)

**Commit Message:** Day 1: Data ingestion complete

## 1. Project Overview & Repository Setup

This capstone project establishes a complete data analytics pipeline for mutual fund schemes in India. The first day focused on creating a standardized project folder layout, configuring environment dependencies, auditing the initial 10 datasets, validating relations, and retrieving real-time data from the official Open API (mfapi.in).

The directory structure is organized as follows:

- **data/raw/**: Holds the 10 raw CSV datasets and newly fetched API CSVs.
- **src/**: Python execution scripts (data\_ingestion.py, live\_nav\_fetch.py).
- **reports/**: Output summaries and generated PDF documentations.
- **requirements.txt**: Stores the package list and a concise data quality summary.

## 2. Dependency Installation

A reproducible environment was configured with package pinning. The main dependencies used include:

- **pandas & numpy**: Core data analysis, clean structured loading, and manipulation.
- **matplotlib, seaborn & plotly**: Visualization libraries for subsequent dashboard features.
- **sqlalchemy**: Database connector interface for importing metrics to SQL databases.
- **requests**: Networking client library to query mfapi.in APIs.
- **scipy**: Analytical calculations and statistical metrics.

- **jupyter**: Interactive environments for draft analytics and investigations.

### 3. Data Ingestion Audit

The script `data_ingestion.py` was executed to read all 10 raw CSV datasets using pandas. Each dataset was audited for shape (rows × columns), column data types, null counts, and duplicates. Below is the audited matrix:

| Filename                     | Description           | Shape       | Null Values Found   |
|------------------------------|-----------------------|-------------|---------------------|
| 01_fund_master.csv           | Fund Master           | 40 x 15     | None                |
| 02_nav_history.csv           | NAV History           | 46,000 x 3  | None                |
| 03_aum_by_fund_house.csv     | AUM by Fund House     | 90 x 5      | None                |
| 04_monthly_sip_inflows.csv   | Monthly SIP Inflows   | 48 x 6      | 12 (yoy_growth_pct) |
| 05_category_inflows.csv      | Category Inflows      | 144 x 3     | None                |
| 06_industry_folio_count.csv  | Industry Folio Count  | 21 x 6      | None                |
| 07_scheme_performance.csv    | Scheme Performance    | 40 x 19     | None                |
| 08_investor_transactions.csv | Investor Transactions | 32,778 x 13 | None                |
| 09_portfolio_holdings.csv    | Portfolio Holdings    | 322 x 8     | None                |
| 10_benchmark_indices.csv     | Benchmark Indices     | 8,050 x 3   | None                |

### 4. Data Quality & Anomalies Report

During audit, only one dataset presented missing values. In **04\_monthly\_sip\_inflows.csv**, the column `yoy_growth_pct` contains 12 null records. An investigation confirmed that these nulls reside within the first 12 chronological rows of the dataset. Because the calculation of Year-over-Year (YoY) growth requires comparative data from the same month in the prior calendar year, no comparison baseline exists for the first year of entries. This is an expected mathematical limitation, rather than data corruption. No duplicates or unparseable dates were found across the other files.

### 5. Fund Master Exploration & AMFI Code Validation

A detailed search on **01\_fund\_master.csv** revealed that the database contains schemes distributed across 10 unique fund houses (e.g. SBI Mutual Fund, HDFC Mutual Fund, ICICI Prudential MF) spanning 2 core categories (Equity and Debt) and 12 distinct sub-categories (e.g. Large Cap, Mid Cap, Liquid, Gilt).

**Validation Analysis:** Every AMFI code (numeric scheme identifier) present in `01_fund_master.csv` was verified against `02_nav_history.csv`. The script confirmed a **100% referential integrity match**: all 40 unique schemes defined in the Fund Master have corresponding historical NAV listings in the NAV History dataset.

## 6. Live API NAV Integration

To ingestion live real-time metrics, the script `live_nav_fetch.py` was written to request recent data from the official endpoint `api.mfapi.in`. The results include:

### A. HDFC Top 100 Direct Plan (AMFI Code: 125497)

Successfully connected to the GET API and retrieved 3,105 rows. The data was parsed from the JSON response, and dates in the response (originally DD-MM-YYYY format) were converted to SQL-compliant standard YYYY-MM-DD format. The parsed records were written to `data/raw/live_hdfc_nav.csv`.

### B. Key Schemes Verification (5 Major Mutual Funds)

Historical and live NAV records were pulled for 5 major schemes to serve as our project benchmarks. The list includes:

- **SBI Bluechip** (AMFI Code: 119551)
- **ICICI Bluechip** (AMFI Code: 120503)
- **Nippon Large Cap** (AMFI Code: 118632)
- **Axis Bluechip** (AMFI Code: 119092)
- **Kotak Bluechip** (AMFI Code: 120841)

The response was parsed, aggregated, and compiled into a single unified dataset, comprising **16,777 combined records**, saved under `data/raw/key_schemes_nav.csv` with correct date formatting.

## 7. Deliverables & Validation Status

All requested Day 1 deliverables have been verified, locally executed, and pushed to your remote repository:

| Deliverable             | Filename / Location                               | Status    |
|-------------------------|---------------------------------------------------|-----------|
| Directory Structure     | data/raw, processed, src, reports, notebooks, sql | COMPLETED |
| Dependencies list       | requirements.txt (with Data Quality summary)      | COMPLETED |
| Ingestion & Exploration | src/data_ingestion.py                             | COMPLETED |
| Live API NAV Loader     | src/live_nav_fetch.py                             | COMPLETED |
| Data Quality Summary    | reports/data_quality_summary.md                   | COMPLETED |
| Ingestion & API Data    | data/raw/live_hdfc_nav.csv & key_schemes_nav.csv  | COMPLETED |
| Version Control Push    | GitHub Repository (Day 1 Commit)                  | COMPLETED |

End of Day 1 Report